HERMES

HERMES

High-Frequency Emergency and Rural Multimedia Exchange System

Email, files, messages and voice over HF radio, where there is no internet

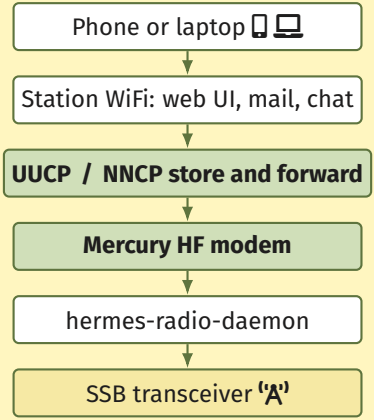
Rhizomatica • **free software** • **November 2026**

github.com/Rhizomatica • hermes.radio



Overview

- A **telecommunication system over HF radio** for communities beyond the reach of the internet and the phone network
- People use it as they use the internet: **email**, webmail and Delta Chat on their phone, over the station's **WiFi**
- The station carries the traffic over **HF** to a gateway, which has an uplink to the internet
- Also **voice**, digital and encrypted where that is permitted, **broadcast** files, and **positions** with an emergency call
- **Free software** from Rhizomatica, supported by ARDC grants since 2021; **hands-on training** so a community runs its own station



Where HERMES runs

Ecuadorian Amazon

- Waorani Indigenous communities: **nearly 90 radios**, the largest deployment
- Some sites are days away on foot or by boat
- Older residents keep in touch with family in other villages

Run by the community

Rhizomatica trains a small group of community members, on site, to install, operate and maintain the station without constant outside support

Bangladesh

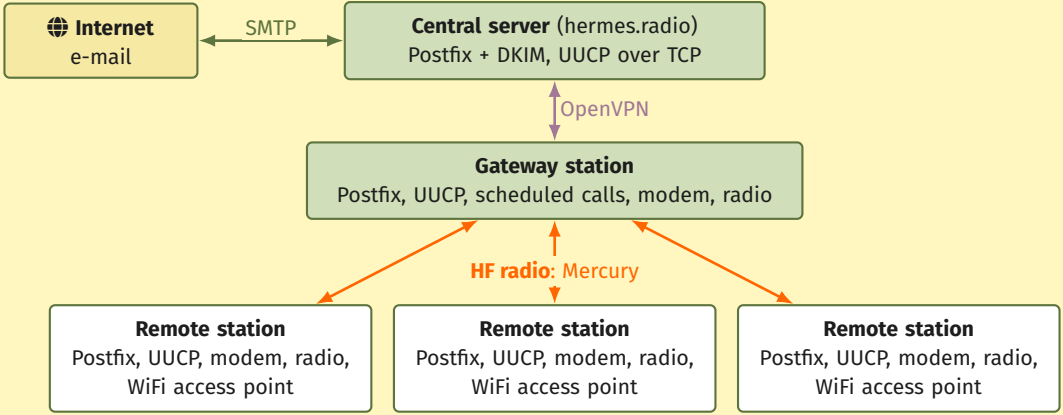
Fishing communities, with HERMES on their **fishing boats**, in a conservation project

Central African Republic

Early warning systems

Source: ARDC, "Rhizomatica's HERMES and Mercury Expand Open Source HF Communications", 6 November 2026

How a network fits together



The central server queues mail for each gateway. The gateway calls the central server over the VPN every few minutes, and its remote stations over HF on schedules set in its web interface.

Source: hermes-installer docs/README.md, docs/deploying-a-network.md

A station

Hardware

- A **Raspberry Pi** (1 to 16 GB) or another Debian computer
- An **SSB transceiver**: the **sBitx** or zBitx, driven directly, or any radio Hamlib controls (IC-7100, IC-7300, FT-710, TS-480, …)
- An antenna, and a WiFi access point for the users
- Optionally a GPS receiver

Software

- Debian 13 and the HERMES package repository
- Postfix and Dovecot, UUCP (or NNCP), Mercury, **hermes-radio-daemon**
- A web interface for messages, users, the radio, the UUCP queue and schedules
- One installer for every station, from a **station file**; a USB installer image for the field

Source: hermes-installer docs/station-files.md; hermes-radio-daemon README.md

Data over HF: store and forward, and broadcast

- **UUCP** carries mail and files over HF by default; it is robust, resumes interrupted transfers, and every station already knows it
- **NNCP**, encrypted and authenticated, is the alternative; a network uses one or the other on every station
- The link is **Mercury**: an ARQ session that delivers a file intact, or reports that it did not
- A station keeps mail while the band is closed and sends it at the next call

Mercury, the link

Free software since May 2026, replacing VARA, a proprietary modem with a paid licence per radio. Seven speed rungs, from MFSK below −10 dB SNR to 3.1 kbit/s; a VARA-compatible TNC. It has its own presentation.

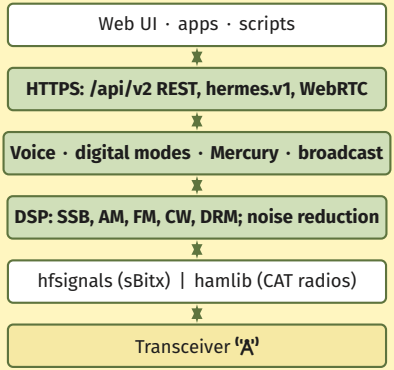
Broadcast: one to many

Files go out as a **RaptorQ fountain**: a receiver that hears enough of *any* frames rebuilds the file, wherever it tuned in. No return path: bulletins and updates to a whole region.

Source: hermes-installer docs/README.md; Mercury docs/presentations/2026-10-mercury; hermes-broadcast WIRE-FORMAT.md

hermes-radio-daemon

- **One daemon** for the radio: control, DSP, audio, voice, digital modes and the modem
- Two backends: **hfsignals** drives the sBitx and zBitx hardware itself (synthesiser, GPIO, codec); **hamlib** drives any CAT radio
- Mercury runs **inside** it, keyed and fed audio directly
- **One front door**, one HTTPS port: the web interface, REST under `/api/v2`, the `hermes.v1` WebSocket, audio over WebRTC
- Spec first: **113 operations** in an OpenAPI contract; Go handlers linked into the daemon; every route needs a login, roles enforced on the server



Source: hermes-radio-daemon README.md; docs/api-design/README.md; api/spec/api-rest.yaml

Digital voice, and encryption where it is permitted

Two digital voice codecs

- **RADE V2**: neural digital voice (FARGAN vocoder, the RADE V2 modem), entirely in C inside the daemon
- **D-STAR**: AMBE voice, compatible with commercial D-STAR radios
- A profile switches between analogue and digital voice

Voice encryption (opt-in)

- For networks where encryption is permitted, **never on amateur bands**
- One 256-bit key shared by the network's stations
- D-STAR: the voice bits XORed with a ChaCha20 keystream
- RADE V2: keyed rotations of the carriers every 40 ms; a late listener locks in about 1 s

On air, October 2026

Both codecs, encrypted and clear, with the right key, the wrong key and none, and a listener joining mid-over: all passed. Listeners heard **no difference** between encrypted and clear speech.

Source: hermes-radio-daemon README.md (Digital Voice, Voice encryption), dsp/rade_voice.h, dsp/dstar_voice.h

Digital text modes and reception reports

- **FT8**: receive and send, a decoder with several passes and signal subtraction
- **JS8**: all four submodes (Normal, Fast, Turbo, Slow), receive and send; free-text messages across several frames
- **CW** and **RTTY**: decoded as **skimmers**, every station in the passband at once
- One log of everything decoded and sent, read through the API
- Received stations reported to **PSKReporter**, with the station's own position from its GPS
- **DRM**: digital broadcast reception, and transmission: music sent and decoded between two stations
- SSB voice processing: equaliser, compressor, limiter; on receive, adaptive noise reduction and AGC

Source: hermes-radio-daemon README.md; dsp/ft8*, dsp/js8*, dsp/pskreporter*

Positions over the air

Position beacons

- A station sends its position on a schedule: from its GPS, or a fixed position typed in
- Every station heard is listed with its last position, and drawn on a **map**
- The beacon format is documented byte by byte

SOS

An emergency code set by a user goes out in every beacon, more often, until the emergency is declared over

Tracks

- The station's own track, and every other station's from its beacons
- A point every few seconds while moving; one every 5 minutes while still
- Fixed **24-byte records**, one file per period: a time range is found by binary search
- A **trip** is a named time range of a track

Source: docs/BEACON-FORMAT.md, gps/gps_track.h, api/spec/api-rest.yaml (tag gps)

Testing and platforms

Testing

- **280 C unit tests** (Unity + fff): DSP, codecs, modes, beacons, tracks
- **Go**: every API handler, its refusals too
- **Integration**: the real daemon on Hamlib's dummy radio
- **Sanitizers** (ASan + UBSan, TSan) and CI on amd64 and arm64 for every pull request
- **On air**: sBitx, IC-7100 and FT-710 stations

Platforms

- **Raspberry Pi** 4 and 5, Debian 13, from the HERMES package repository; Mercury even on a Pi Zero, with the zBitx
- **A phone as the station**: the daemon and Mercury on Sailfish OS and a PinePhone Pro, the radio on USB (CAT without kernel drivers, proposed to Hamlib), the phone's own GPS
- Mercury also on **Android** and **iOS** phones, **macOS** and **Windows**; iOS gives apps no USB access

Source: tests/, .github/workflows; hermes-radio-daemon README.md; Hamlib pull request 2272

2026, and what comes next

580 commits to the radio daemon since 25 April

117 pull requests merged in it

1468 commits to Mercury since its C rewrite

12 stations in a network installed in November

⇄ Store and forward

UUCP and NNCP behind one layer of the station API; SOS messages

One interface

The web interfaces on the station API; the old WebSocket retired

Sharing the channel

HF channel access between stations; Mercury's optional encryption

Other networks

Reticulum over HF (done); Meshtastic, for LoRa networks (in progress)

Source: git history of hermes-radio-daemon and Mercury; ARDC, 6 November 2026

HERMES

THANK YOU

github.com/Rhizomatica

hermes.radio • rhizomatica.org/hermes

lists.riseup.net/www/info/hermes-general

HERMES is free software.

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